

Project AeroVision

Product Requirements Document (PRD)

1. Executive Summary

Air pollution remains a critical public health hazard, responsible for millions of excess deaths annually. While the Air Quality Index (AQI) effectively communicates pollution levels into a single number, relying solely on ground stations is insufficient as most of the global population lives over 100 km away from them. Furthermore, Formaldehyde (HCHO) acts as a crucial indicator of volatile organic compounds (VOCs) and tropospheric ozone formation, particularly during biomass burning events like agricultural fires.

Project AeroVision will develop satellite-derived spatial models to bridge this monitoring gap. The project directly aligns with the National Clean Air Programme (NCAP) targets, UN Sustainable Development Goal (SDG) 11.6, and ISRO/SAC mandates for environmental monitoring.

2. Project Objectives

- **Objective 1: Surface AQI Modeling:** Produce daily spatial maps of surface AQI across India by developing a hybrid CNN-LSTM deep learning model trained on satellite and ground observations.
- **Objective 2: HCHO Hotspot Detection:** Generate high-resolution HCHO hotspot maps during distinct biomass burning seasons, pinpoint major source regions (e.g., Indo-Gangetic Plain, forest fire zones), and analyze fire-HCHO transport pathways.

3. Data Requirements

All modeling steps require rigorous temporal co-registration, spatial resampling, and quality-flag filtering.

Dataset / Mission	Source Portal	Key Variables
INSAT-3D	MOSDAC	AOD (550 nm)
Sentinel-5P	GEE / DLR	NO2, SO2, CO, O3, HCHO
CPCB Ground	CPCB API	PM2.5, PM10, Criteria Pollutants
MODIS / VIIRS	NASA FIRMS	Fire Radiative Power (FRP)
ERA5 / IMDAA	Copernicus	Winds, Temp, RH, PBLH

4. Phased Execution Strategy

Phase 1: Data Acquisition & Preprocessing (Weeks 1–3)

- **Harmonization:** Download all datasets and regrid them to daily means on a common $0.1^{\circ} \times 0.1^{\circ}$ spatial grid using xarray.
- **Quality Control:** Remove satellite pixels with a cloud fraction > 0.3 and a TROPOMI QA flag < 0.75 .
- **Data Splitting:** Divide data by geographic region into 70% training, 15% validation, and 15% testing splits to prevent spatial leakage.

Phase 2: Surface AQI Modelling (Weeks 4–9)

- **Architecture:** Deploy a CNN-LSTM deep learning model. 2D CNN layers will extract spatial pollution patterns, which are then passed into an LSTM network to learn multi-day temporal dependencies.
- **Training & Tuning:** Pre-train using data from high-density monitoring regions (NCR, Mumbai), followed by fine-tuning on sparse regions. Utilize Bayesian optimization for hyperparameter tuning.
- **Standardization:** Convert predicted surface concentrations directly into India's National AQI (NAQI / CPCB) categories.
- **Maintenance:** Retrain the model monthly to account for seasonal drifts in the AOD–PM_{2.5} relationship.

Phase 3: HCHO Hotspot Analysis (Weeks 6–11)

- **Seasonality:** Extract daily FRP data during key biomass burning seasons: Kharif (Oct–Nov), Rabi (Apr–May), and Forest Fires (Mar–May).
- **Detection Methodology:** Flag pixels exceeding the 95th percentile of the multi-year HCHO climatology. Group nearby elevated pixels using DBSCAN spatial clustering (epsilon = 50 km, min-samples = 5).
- **Transport Analysis:** Use NOAA HYSPLIT back-trajectory models and ERA5 850 hPa wind vectors to map dominant transport pathways.

Phase 4: Validation & Dissemination (Weeks 10–13)

- **Outputs:** Produce 0.1° resolution daily/seasonal-mean AQI maps and multi-year HCHO anomaly maps.
- **Key Deliverables:**
 - Technical methodology and results report.
 - GIS shapefile/NetCDF data packages.
 - Interactive web maps via Folium.
 - Peer-reviewed journal manuscript.

5. Technology Stack

- **Data Access:** Google Earth Engine, MOSDAC, CPCB API, FIRMS, Copernicus CDS.
- **Data Processing:** Python, xarray, netCDF4, NumPy, Pandas, SciPy.
- **Deep Learning:** TensorFlow / PyTorch, scikit-learn.
- **Visualisation:** Matplotlib, Cartopy, GeoPandas, Folium, Plotly.

6. Success Metrics & Evaluation

Feature	Key Evaluation Metrics
AQI Model	RMSE, MAE, and spatial bias per pollutant species.
Correlation	Pearson R correlation against CPCB observations.
Classification	Confusion matrix assessing NAQI category accuracy.
HCHO Hotspots	Recall and detection accuracy of hotspot regions.
Analysis	Fire-HCHO correlation strength (r-value, p-value).

Approval Signatures

Project Lead:

Person

Date:

Date